ECO 319/619

Time Series Analysis and Forecasting
R Code

Topic - Effects of EV (Electric Vehicle) sales on Gasoline Prices in India

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Research Problem:

How the sales of EV (Electric Vehicles) affect the Gasoline Prices in India?

Variables Used:

The data span ranges from January 2015 to June 2025. The variables used for the above study includes:

1. Electric Vehicles sales in India
2. Gasoline Prices (INR)
3. Exchange rate (LAV in %)
4. CPI
5. Brent Crude Oil Price (INR)

Installing required packages

```
> library(readxl)
> EVSales_dataset <- read_excel("EV_Sales/EVSales_dataset.xlsx")
> View(EVSales_dataset)
> attach(EVSales_dataset)
> install.packages(c("dplyr", "ggplot2", "vars", "forecast", "tseries",
"lmtest", "urca", "readxl"))
Error in install.packages : Updating loaded packages


Restarting R session...

> install.packages(c("dplyr", "ggplot2", "vars", "forecast", "tseries",
"lmtest", "urca", "readxl"))

WARNING: Rtools is required to build R packages but is not currently installed.
Please download and install the appropriate version of Rtools before proceeding:


https://cran.rstudio.com/bin/windows/Rtools/
Installing packages into 'C:/Users/Aditi/AppData/Local/R/win-library/4.5'
(as 'lib' is unspecified)
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/dplyr_1.1.4.zip'
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/ggplot2_4.0.0.zip'
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/vars_1.6-1.zip'
trying URL
'https://cran.rstudio.com/bin/windows/contrib/4.5/forecast_8.24.0.zip'
```

```
trying URL
'https://cran.rstudio.com/bin/windows/contrib/4.5/tseries_0.10-58.zip'
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/lmtest_0.9-40.zip'
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/urca_1.3-4.zip'
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/readxl_1.4.5.zip'
package ‘dplyr’ successfully unpacked and MD5 sums checked
package ‘ggplot2’ successfully unpacked and MD5 sums checked
package ‘vars’ successfully unpacked and MD5 sums checked
package ‘forecast’ successfully unpacked and MD5 sums checked
package ‘tseries’ successfully unpacked and MD5 sums checked
package ‘lmtest’ successfully unpacked and MD5 sums checked
package ‘urca’ successfully unpacked and MD5 sums checked
package ‘readxl’ successfully unpacked and MD5 sums checked

The downloaded binary packages are in
 C:\Users\Aditi\AppData\Local\Temp\RtmpcpxWHt\downloaded_packages
> library(dplyr)


Attaching package: ‘dplyr’

The following objects are masked from ‘package:stats’:

    filter, lag

The following objects are masked from ‘package:base’:

    intersect, setdiff, setequal, union
> library(ggplot2)
> library(vars)

Loading required package: MASS

Attaching package: ‘MASS’

The following object is masked from ‘package:dplyr’:
```

```
    select

Loading required package: strucchange
Loading required package: zoo

Attaching package: ‘zoo’

The following objects are masked from ‘package:base’:

    as.Date, as.Date.numeric

Loading required package: sandwich
Loading required package: urca
Loading required package: lmtest
 > library(forecast)

Registered S3 method overwritten by 'quantmod':
  method            from
  as.zoo.data.frame zoo
This is forecast 8.24.0
  Need help getting started? Try the online textbook FPP:
 http://otexts.com/fpp2/
> library(tseries)


   ‘tseries’ version: 0.10-58

    ‘tseries’ is a package for time series analysis and computational finance.

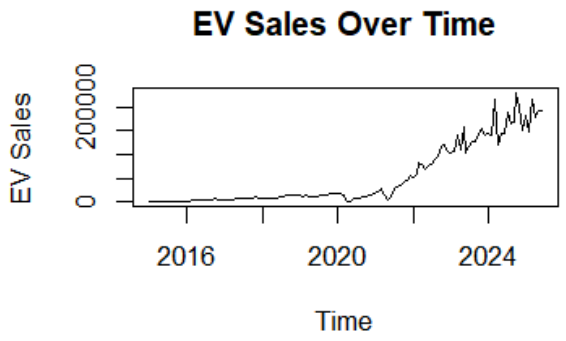
    See ‘library(help="tseries")’ for details.
> library(lmtest)
> library(urca)
> library(readxl)
> ev_data  <- read_excel("EV_Sales/EVSales_dataset.xlsx")
```

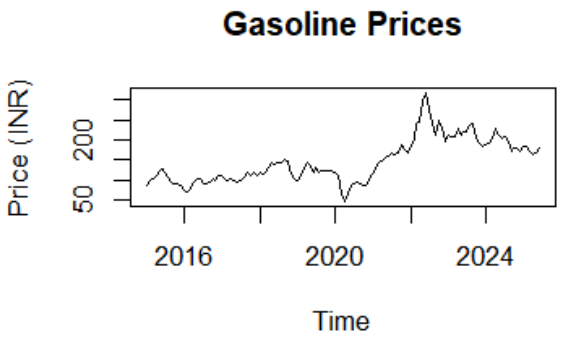
```
> str(ev_data)
tibble [126 × 6] (S3: tbl_df/tbl/data.frame)
 $ YEAR                       : chr [1:126] "2015-01" "2015-02" "2015-03"
"2015-04" ...
 $ TOTAL_EV                   : num [1:126] 179 177 215 168 194 ...
 $ GASOLINE PRICES (INR)      : num [1:126] 84.8 99.7 102.7 112.5 123.7 ...
 $ EXCHANGE RATE (LAV IN %)   : num [1:126] 12.5 12.5 12.5 12.5 12.5 ...
 $ BRENT CRUDE OIL PRICE(INR): num [1:126] 4217 5492 4764 5670 5605 ...
 $ CPI                        : num [1:126] 120 120 120 121 122 ...
> summary(ev_data)
     YEAR              TOTAL_EV      GASOLINE PRICES (INR) EXCHANGE RATE (LAV IN
%)
 Length:126         Min.   :   168   Min.   : 45.18        Min.   :12.14
 Class :character   1st Qu.:  6631   1st Qu.:102.64        1st Qu.:12.47
 Mode  :character   Median : 14089   Median :128.09        Median :12.49
                    Mean   : 52123   Mean   :146.68        Mean   :12.45
                    3rd Qu.:102830   3rd Qu.:185.64        3rd Qu.:12.49
                    Max.   :229673   Max.   :318.72        Max.   :12.49
                   BRENT CRUDE OIL PRICE(INR)      CPI
                    Min.   : 1318                 Min.   :119.5
                    1st Qu.: 4618                 1st Qu.:134.0
                    Median : 5924                 Median :150.7
                    Mean   : 5909                 Mean   :154.6
                    3rd Qu.: 6903                 3rd Qu.:175.7
                    Max.   :11140                 Max.   :196.5
> ev_data$DATE <- as.Date(paste0(ev_data$YEAR, "-01"), format = "%Y-%m-%d")
> ts_data <- ts(ev_data[, c("TOTAL_EV", "GASOLINE PRICES (INR)", "EXCHANGE RATE
(LAV IN %)",
+                           "BRENT CRUDE OIL PRICE(INR)", "CPI")],
+               start = c(2015, 1), frequency = 12)

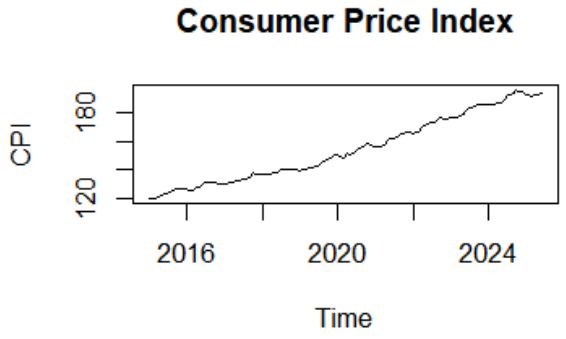
 > summary(ts_data)
 TOTAL_EV      GASOLINE PRICES (INR) EXCHANGE RATE (LAV IN %) BRENT CRUDE OIL
                                                                PRICE(INR)
 Min.   :   168   Min.   : 45.18        Min.   :12.14           Min.   : 1318
 1st Qu.:  6631   1st Qu.:102.64        1st Qu.:12.47           1st Qu.: 4618
```

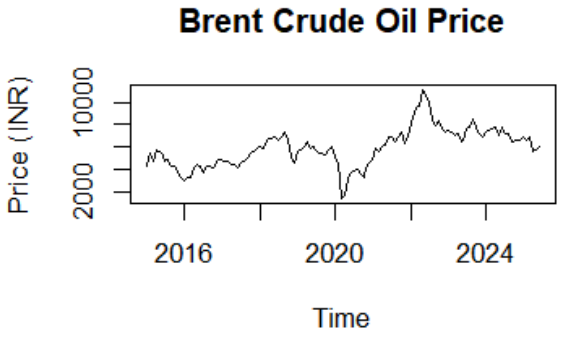
```
 Median : 14089   Median :128.09        Median :12.49            Median : 5924
 Mean   : 52123   Mean   :146.68        Mean   :12.45            Mean   : 5909
 3rd Qu.:102830   3rd Qu.:185.64        3rd Qu.:12.49            3rd Qu.: 6903
 Max.   :229673   Max.   :318.72        Max.   :12.49            Max.   :11140
      CPI
 Min.   :119.5
 1st Qu.:134.0
 Median :150.7
 Mean   :154.6
 3rd Qu.:175.7
 Max.   :196.5
```

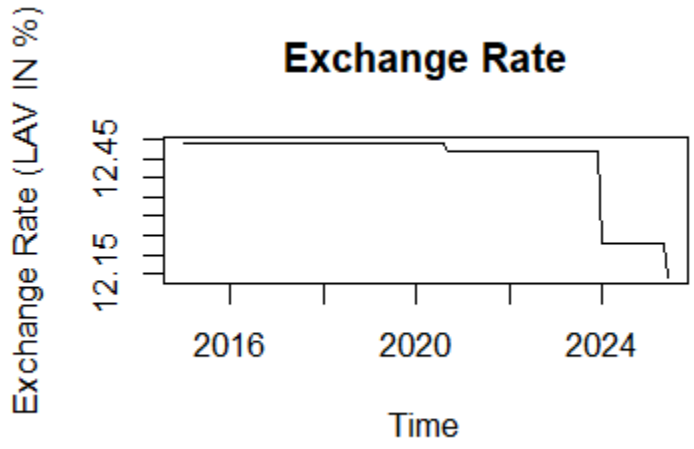
```
> par(mfrow = c(2, 2))
> plot(ts_data[, "TOTAL_EV"], main = "EV Sales Over Time", ylab = "EV Sales")
> plot(ts_data[, "GASOLINE PRICES (INR)"], main = "Gasoline Prices", ylab =
"Price (INR)")
>
> plot(ts_data[, "CPI"], main = "Consumer Price Index", ylab = "CPI")
> plot(ts_data[, "BRENT CRUDE OIL PRICE(INR)"], main = "Brent Crude Oil Price",
ylab = "Price (INR)")
> plot(ts_data[, "EXCHANGE RATE (LAV IN %)"], main = "Exchange Rate", ylab =
"Exchange Rate (LAV IN %)")
```











Extracting year and month from DATE

```
> ev_data$YEAR_NUM <- as.numeric(format(ev_data$DATE, "%Y"))
> ev_data$MONTH_NUM <- as.numeric(format(ev_data$DATE, "%m"))
```

FAME 1 dummy (April 2015 to March 2019)

```
> ev_data$FAME1 <- ifelse(ev_data$DATE >= as.Date("2015-04-01") &
```

```
+                             ev_data$DATE <= as.Date("2019-03-31"), 1, 0)
```

FAME 2 dummy (April 2019 to March 2024)

```
> ev_data$FAME2 <- ifelse(ev_data$DATE >= as.Date("2019-04-01") &
+                             ev_data$DATE <= as.Date("2024-03-31"), 1, 0)
```

COVID Effect dummy (March 2020 onwards)

```
> ev_data$COVID <- ifelse(ev_data$DATE >= as.Date("2020-03-01"), 1, 0)
```

Russia-Ukraine war dummy (From February 2022)

```
> ev_data$RUS_UKR_WAR <- ifelse(ev_data$DATE >= as.Date("2022-02-01"), 1, 0)
> dummies <- ts(ev_data[, c("FAME1", "FAME2", "COVID", "RUS_UKR_WAR")],
+               start = c(start_year, start_month), frequency = 12)
> summary(dummies)
```

```
     FAME1            FAME2             COVID          RUS_UKR_WAR
 Min.   :0.000   Min.   :0.0000   Min.   :0.0000   Min.   :0.0000
 1st Qu.:0.000   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.0000
 Median :0.000   Median :0.0000   Median :1.0000   Median :0.0000
 Mean   :0.381   Mean   :0.4762   Mean   :0.5079   Mean   :0.3254
 3rd Qu.:1.000   3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.:1.0000
 Max.   :1.000   Max.   :1.0000   Max.   :1.0000   Max.   :1.0000
```

```
> cor_matrix <- cor(ts_data)
> print("Correlation Matrix:")
[1] "Correlation Matrix:"
> print(cor_matrix)
```

```
                          TOTAL_EV GASOLINE PRICES (INR) EXCHANGE RATE (LAV IN %)
TOTAL_EV                 1.0000000             0.7116028               -0.8063883
GASOLINE PRICES (INR)    0.7116028             1.0000000               -0.3802560
EXCHANGE RATE (LAV IN %) -0.8063883            -0.3802560                1.0000000
BRENT CRUDE OIL PRICE(INR) 0.5537372            0.9290346               -0.2776099
CPI                      0.9072351             0.7631661               -0.7066925
                         BRENT CRUDE OIL PRICE(INR)        CPI
TOTAL_EV                                  0.5537372  0.9072351
GASOLINE PRICES (INR)                     0.9290346  0.7631661
EXCHANGE RATE (LAV IN %)                 -0.2776099 -0.7066925
BRENT CRUDE OIL PRICE(INR)                1.0000000  0.6107497
CPI                                       0.6107497  1.0000000
```

Checking for Stationarity using ADF test

```
> adf_results <- list()
> for(i in 1:ncol(ts_data)) {
+   adf_results[[colnames(ts_data)[i]]] <- adf.test(ts_data[, i])
+ }
>
> print(adf_results)
$TOTAL_EV
```

```
        Augmented Dickey-Fuller Test

data:  ts_data[, i]
Dickey-Fuller = -0.81545, Lag order = 4, p-value = 0.9578
alternative hypothesis: stationary


$`GASOLINE PRICES (INR)`

        Augmented Dickey-Fuller Test

data:  ts_data[, i]
Dickey-Fuller = -2.3184, Lag order = 4, p-value = 0.4442
alternative hypothesis: stationary


$`EXCHANGE RATE (LAV IN %)`

        Augmented Dickey-Fuller Test

data:  ts_data[, i]
Dickey-Fuller = -0.55417, Lag order = 4, p-value = 0.9781
alternative hypothesis: stationary


$`BRENT CRUDE OIL PRICE(INR)`

        Augmented Dickey-Fuller Test

data:  ts_data[, i]
Dickey-Fuller = -2.4429, Lag order = 4, p-value = 0.3925
alternative hypothesis: stationary


$CPI

        Augmented Dickey-Fuller Test

data:  ts_data[, i]
Dickey-Fuller = -1.9923, Lag order = 4, p-value = 0.5797
alternative hypothesis: stationary

> ts_data_diff <- diff(ts_data)
> adf_results_diff <- list()
> for(i in 1:ncol(ts_data_diff)) {
+     adf_results_diff[[colnames(ts_data_diff)[i]]] <- adf.test(ts_data_diff[,
i])
+ }
Warning messages:
1: In adf.test(ts_data_diff[, i]) : p-value smaller than printed p-value
2: In adf.test(ts_data_diff[, i]) : p-value smaller than printed p-value
3: In adf.test(ts_data_diff[, i]) : p-value smaller than printed p-value
4: In adf.test(ts_data_diff[, i]) : p-value smaller than printed p-value
5: In adf.test(ts_data_diff[, i]) : p-value smaller than printed p-value
> print(adf_results_diff)

$TOTAL_EV
```

```
 Augmented Dickey-Fuller Test

data:  ts_data_diff[, i]
Dickey-Fuller = -6.9482, Lag order
= 4, p-value = 0.01
alternative hypothesis: stationary


$`GASOLINE PRICES (INR)`

 Augmented Dickey-Fuller Test

data:  ts_data_diff[, i]
Dickey-Fuller = -5.3138, Lag order
= 4, p-value = 0.01
alternative hypothesis: stationary


$`EXCHANGE RATE (LAV IN %)`

 Augmented Dickey-Fuller Test

data:  ts_data_diff[, i]
Dickey-Fuller = -5.0947, Lag order
= 4, p-value = 0.01
alternative hypothesis: stationary


$`BRENT CRUDE OIL PRICE(INR)`

 Augmented Dickey-Fuller Test

data:  ts_data_diff[, i]
Dickey-Fuller = -5.4374, Lag order
```

```
= 4, p-value = 0.01
alternative hypothesis: stationary


$CPI


 Augmented Dickey-Fuller Test


data:  ts_data_diff[, i]
Dickey-Fuller = -6.7494, Lag order
= 4, p-value = 0.01
alternative hypothesis: stationary
```

As residuals are not independent over time, there exists a serial correlation in data. Also skewness shows significant asymmetry and kurtosis show extreme values (heavy tails). Therefore we use Bayesian VAR which treats model parameters as random with probability distribution instead of normal VAR which treats parameters as fixed values.

```
> install.packages("BVAR")

 WARNING: Rtools is required to build R packages but is not currently installed.
 Please download and install the appropriate version of Rtools before
 proceeding:

 https://cran.rstudio.com/bin/windows/Rtools/


 Installing package into ‘C:/Users/Aditi/AppData/Local/R/win-library/4.5’
(as ‘lib’ is unspecified)

 trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/BVAR_1.0.5.zip'
 Content type 'application/zip' length 1198935 bytes (1.1 MB)
 downloaded 1.1 MB

 package ‘BVAR’ successfully unpacked and MD5 sums checked

 The downloaded binary packages are in
       C:\Users\Aditi\AppData\Local\Temp\RtmpsZFB5f\downloaded_packages
> library(BVAR)


 Attaching package: ‘BVAR’
```

```
The following objects are masked from 'package:vars':

    fevd, irf

> dummy_matrix <- as.matrix(ev_data[, c("FAME1", "FAME2", "COVID",
+                                       "RUS_UKR_WAR")])
> bvar_with_dummies <- bvar(
+     ts_data_diff,
+     lags = 4,
+     exogen = dummy_matrix,
+     priors = bv_priors(hyper = "auto")
+     )
Optimisation concluded.
Posterior marginal likelihood: -2928.548
Hyperparameters: lambda = 0.009

|=========================================================================
=| 100%
Finished MCMC after 5.25 secs.
```

```
> summary(bvar_with_dummies)
Bayesian VAR consisting of 121 observations, 5 variables and 4 lags.
Time spent calculating: 2.46 secs
Hyperparameters: lambda
Hyperparameter values after optimisation: 0.009
Iterations (burnt / thinning): 10000 (5000 / 1)
Accepted draws (rate): 771 (0.154)

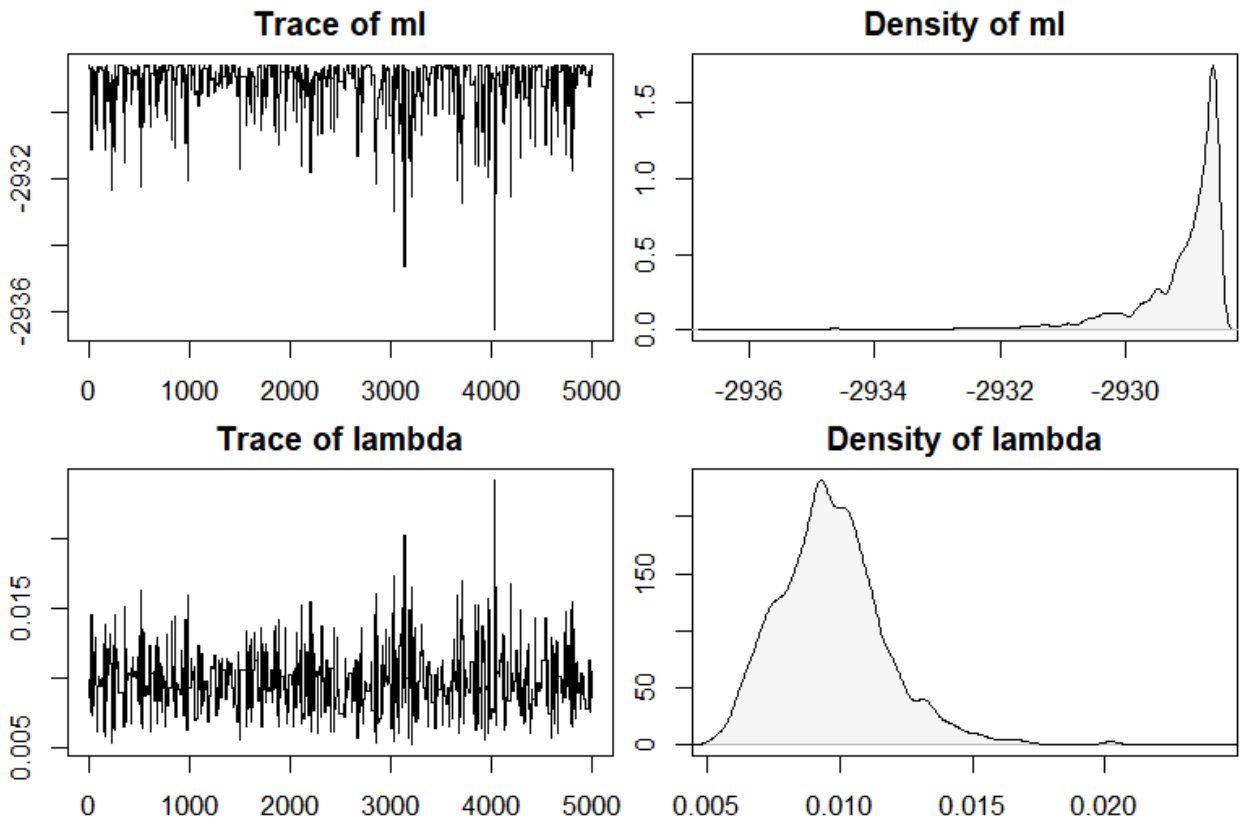
Numeric array (dimensions 21, 5) of coefficient values from a BVAR.
Median values:
                                TOTAL_EV GASOLINE PRICES (INR) EXCHANGE RATE (LAV IN %)
constant                        3978.848                -0.175                        0
TOTAL_EV-lag1                     -0.735                 0.000                        0
GASOLINE PRICES (INR)-lag1        -8.187                 0.916                        0
EXCHANGE RATE (LAV IN %)-lag1     14.868                -0.002                        1
BRENT CRUDE OIL PRICE(INR)-lag1    0.823                -0.002                        0
CPI-lag1                          32.274                 0.008                        0
TOTAL_EV-lag2                     -0.412                 0.000                        0
GASOLINE PRICES (INR)-lag2        -3.894                -0.006                        0
EXCHANGE RATE (LAV IN %)-lag2     -8.010                 0.003                        0
BRENT CRUDE OIL PRICE(INR)-lag2    0.772                -0.006                        0
CPI-lag2                          -3.519                -0.008                        0
TOTAL_EV-lag3                     -0.310                 0.000                        0
GASOLINE PRICES (INR)-lag3        -1.195                 0.001                        0
EXCHANGE RATE (LAV IN %)-lag3      4.076                -0.005                        0
BRENT CRUDE OIL PRICE(INR)-lag3   -0.191                 0.000                        0
CPI-lag3                           0.808                -0.003                        0
TOTAL_EV-lag4                     -0.082                 0.000                        0
GASOLINE PRICES (INR)-lag4         0.568                 0.000                        0
EXCHANGE RATE (LAV IN %)-lag4     -3.179                 0.004                        0
BRENT CRUDE OIL PRICE(INR)-lag4    0.263                 0.000                        0
CPI-lag4                           1.823                -0.001                        0
```

```
                                BRENT CRUDE OIL PRICE(INR)   CPI
constant                                             1.398 0.006
TOTAL_EV-lag1                                       -0.002 0.000
GASOLINE PRICES (INR)-lag1                          -0.661 0.000
EXCHANGE RATE (LAV IN %)-lag1                       -1.008 0.000
BRENT CRUDE OIL PRICE(INR)-lag1                      0.264 0.000
CPI-lag1                                             0.834 0.992
TOTAL_EV-lag2                                       -0.002 0.000
GASOLINE PRICES (INR)-lag2                          -0.078 0.000
EXCHANGE RATE (LAV IN %)-lag2                        0.152 0.000
BRENT CRUDE OIL PRICE(INR)-lag2                     -0.051 0.000
CPI-lag2                                            -0.172 0.000
TOTAL_EV-lag3                                        0.002 0.000
GASOLINE PRICES (INR)-lag3                          -0.036 0.000
EXCHANGE RATE (LAV IN %)-lag3                       -0.435 0.000
BRENT CRUDE OIL PRICE(INR)-lag3                     -0.044 0.000
CPI-lag3                                            -0.055 0.000
TOTAL_EV-lag4                                        0.002 0.000
GASOLINE PRICES (INR)-lag4                          -0.013 0.000
EXCHANGE RATE (LAV IN %)-lag4                        0.078 0.000
BRENT CRUDE OIL PRICE(INR)-lag4                     -0.016 0.000
CPI-lag4                                            -0.017 0.000

Numeric array (dimensions 5, 5) of variance-covariance values from a BVAR.
Median values:
                            TOTAL_EV GASOLINE PRICES (INR) EXCHANGE RATE (LAV IN %)
TOTAL_EV                220474234.531             10112.664                   -16.438
GASOLINE PRICES (INR)       10112.664               286.512                     0.001
EXCHANGE RATE (LAV IN %)      -16.438                 0.001                     0.001
BRENT CRUDE OIL PRICE(INR)  -769865.673              5602.155                    -1.227
CPI                           -1901.965                -0.225                     0.000
                        BRENT CRUDE OIL PRICE(INR)       CPI
TOTAL_EV                          -769865.673 -1901.965
GASOLINE PRICES (INR)                5602.155    -0.225
EXCHANGE RATE (LAV IN %)               -1.227     0.000
BRENT CRUDE OIL PRICE(INR)         391500.224   -22.430
CPI                                   -22.430     1.274

Log-Likelihood: -2707.359
```

```
> plot(bvar_with_dummies)
```




```
> complete_beta <- bvar_with_dummies$beta
> median_coefs <- apply(complete_beta, 2:3, median)
> cat("Raw coefficient matrix dimensions:", dim(median_coefs), "\n\n")
Raw coefficient matrix dimensions: 21 5

> for (i in 1:nrow(median_coefs)) {
+     cat("Row", i, ":", paste(round(median_coefs[i, ], 4), collapse = ", "),
+         "\n")
+ }
Row 1 : 3975.9372, -0.1498, -4e-04, 2.7151, 0.0079
Row 2 : -0.7338, 0, 0, -0.0017, 0
Row 3 : -8.3901, 0.9166, 0, -0.6772, 2e-04
Row 4 : -18.9219, -0.0055, 0.9997, 0.0338, 1e-04
Row 5 : 0.9032, -0.002, 0, 0.2703, -1e-04
Row 6 : 31.723, 0.0129, 0, 1.0654, 0.992
Row 7 : -0.4132, -1e-04, 0, -0.0015, 0
Row 8 : -3.1674, -0.0065, 0, -0.055, 1e-04
Row 9 : -19.7657, 0.0031, 0, -0.0908, -3e-04
Row 10 : 0.7632, -0.0064, 0, -0.051, 1e-04
Row 11 : -2.0781, -0.0108, 0, -0.2381, -4e-04
Row 12 : -0.3099, 1e-04, 0, 0.0025, 0
Row 13 : -1.0917, 0.0018, 0, -0.0195, 1e-04
Row 14 : 5.8841, 8e-04, 0, 0.1208, 0
Row 15 : -0.2024, -3e-04, 0, -0.043, 0
Row 16 : 4.9424, -0.0016, 0, -0.1, 0
Row 17 : -0.0788, 1e-04, 0, 0.0021, 0
Row 18 : 0.2866, -6e-04, 0, -0.0099, -1e-04
```

```
Row 19 : 5.7971, 0.0106, 0, 0.244, 5e-04
Row 20 : 0.2827, -3e-04, 0, -0.0178, 0
Row 21 : 0.9045, 0, 0, -0.0184, -2e-04
```

```
> n_total_rows <- nrow(median_coefs)
> dummy_rows <- (n_total_rows - 3):n_total_rows
> cat("\nLast 4 rows (likely your dummy variables):\n")
```

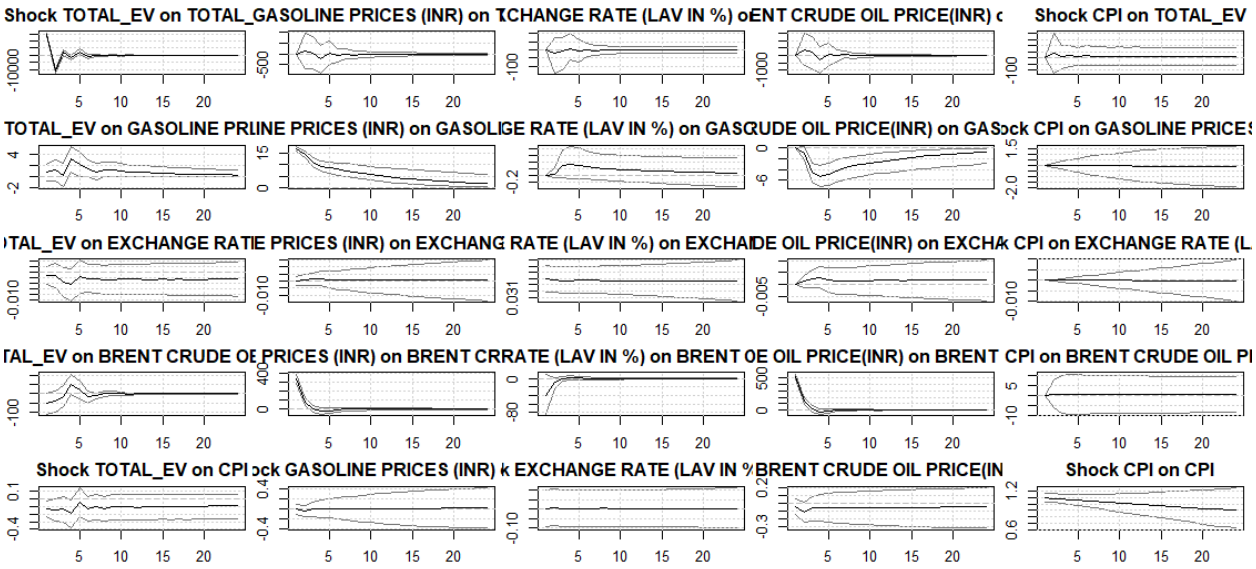
```
Last 4 rows (likely your dummy variables):
```

```
> dummy_effects <- median_coefs[dummy_rows, ]
> rownames(dummy_effects) <- colnames(dummy_matrix)
> colnames(dummy_effects) <- colnames(ts_data_diff)
```

```
> print(dummy_effects)
             TOTAL_EV GASOLINE PRICES (INR) EXCHANGE RATE (LAV IN %)
FAME1       0.6970918          -0.0006374465              -1.812318e-06
FAME2       4.8978007          -0.0037921430               8.868100e-06
COVID       0.2372271          -0.0002512031              -2.201474e-06
RUS_UKR_WAR 0.3408273          -0.0011162594              -9.024258e-07
            BRENT CRUDE OIL PRICE(INR)           CPI
FAME1                      -0.02023929 -7.201771e-05
FAME2                       0.10104699 -2.032445e-04
COVID                      -0.01462926 -7.795452e-07
RUS_UKR_WAR                -0.06380327 -2.480987e-04
```

Impulse Response Function:

```
> library(BVAR)
> irf_results <- irf(bvar_with_dummies, horizon = 24)
> plot(irf_results)
```



Granger Causality Test:

```
> library(vars)


> TOTAL_EV_ts <- ts(ev_data$TOTAL_EV, start = c(2015, 1), frequency = 12)
> GASOLINE_ts <- ts(ev_data$`GASOLINE PRICES (INR)`, start = c(2015, 1),
frequency = 12)
> bivariate_ts <- cbind(TOTAL_EV_ts, GASOLINE_ts)
> colnames(bivariate_ts) <- c("TOTAL_EV", "GASOLINE_PRICES")
> var_model <- VAR(bivariate_ts, p = 4)
> cat("=== GRANGER CAUSALITY RESULTS ===\n\n")
=== GRANGER CAUSALITY RESULTS ===

> cat("Gasoline Prices -> EV Sales:\n")
Gasoline Prices -> EV Sales:
> test1 <- causality(var_model, cause = "GASOLINE_PRICES")
> print(test1$Granger)

      Granger causality H0: GASOLINE_PRICES do not Granger-cause TOTAL_EV

data:  VAR object var_model
F-Test = 2.5152, df1 = 4, df2 = 226, p-value = 0.04238
> cat("Conclusion: ", ifelse(test1$Granger$p.value < 0.05,
+ "SIGNIFICANT - Gasoline prices Granger-cause EV sales",
+ "NOT SIGNIFICANT - No Granger causality"), "\n\n")
Conclusion:  SIGNIFICANT - Gasoline prices Granger-cause EV sales

> cat("EV Sales -> Gasoline Prices:\n")
EV Sales -> Gasoline Prices:
> test2 <- causality(var_model, cause = "TOTAL_EV")
> print(test2$Granger)

      Granger causality H0: TOTAL_EV do not Granger-cause GASOLINE_PRICES

data:  VAR object var_model
F-Test = 0.96017, df1 = 4, df2 = 226, p-value = 0.4302

> cat("Conclusion: ", ifelse(test2$Granger$p.value < 0.05,
+                            "SIGNIFICANT - EV sales Granger-cause gasoline
prices",
+                            "NOT SIGNIFICANT - No Granger causality"), "\n")
Conclusion:  NOT SIGNIFICANT - No Granger causality
```

FEVD:

```
> library(vars)
> var_model <- VAR(ts_data_diff, p = 4, type = "const")
> fevd_result <- fevd(var_model, n.ahead = 12)
> print(fevd_result)
$TOTAL_EV
       TOTAL_EV GASOLINE.PRICES..INR. EXCHANGE.RATE..LAV.IN...
 [1,] 1.0000000          0.0000000000             0.000000e+00
 [2,] 0.9489144          0.0004639577             7.313008e-05
 [3,] 0.7667711          0.0122800966             1.429483e-01
 [4,] 0.5727474          0.0098043140             3.272895e-01
```

```
 [5,] 0.5632661          0.0105882652            3.306191e-01
 [6,] 0.5580158          0.0114569803            3.296087e-01
 [7,] 0.5507861          0.0243084840            3.259403e-01
 [8,] 0.5482619          0.0282352287            3.233072e-01
 [9,] 0.5445025          0.0285008032            3.247302e-01
[10,] 0.5401575          0.0291397004            3.285449e-01
[11,] 0.5392085          0.0293334720            3.287999e-01
[12,] 0.5389844          0.0293166383            3.287171e-01
      BRENT.CRUDE.OIL.PRICE.INR.        CPI
 [1,]                0.000000000 0.00000000
 [2,]                0.011804666 0.03874384
 [3,]                0.009728558 0.06827192
 [4,]                0.007378336 0.08278047
 [5,]                0.010664153 0.08486241
 [6,]                0.013742463 0.08717607
 [7,]                0.013917189 0.08504789
 [8,]                0.015771400 0.08442421
 [9,]                0.018128900 0.08413758
[10,]                0.018145811 0.08401216
[11,]                0.018840354 0.08381773
[12,]                0.019269180 0.08371269

$GASOLINE.PRICES..INR.
         TOTAL_EV GASOLINE.PRICES..INR. EXCHANGE.RATE..LAV.IN...
 [1,] 0.005882618             0.9941174             0.0000000000
 [2,] 0.005252080             0.7099589             0.0001329722
 [3,] 0.005069532             0.6978274             0.0019312386
 [4,] 0.023551802             0.6523237             0.0148705118
 [5,] 0.022881503             0.6236557             0.0345506493
 [6,] 0.033498212             0.6112217             0.0337561285
 [7,] 0.033948762             0.6118855             0.0338064884
 [8,] 0.033732464             0.6084672             0.0368289480
 [9,] 0.033522429             0.6032140             0.0367332183
[10,] 0.033397150             0.5993478             0.0369610032
[11,] 0.033778095             0.5987412             0.0370186357
[12,] 0.034392385             0.5982970             0.0370782148
      BRENT.CRUDE.OIL.PRICE.INR.        CPI
 [1,]                 0.0000000 0.00000000
 [2,]                 0.2306154 0.05404062
 [3,]                 0.2378998 0.05727204
 [4,]                 0.2259104 0.08334355
 [5,]                 0.2167297 0.10218253
 [6,]                 0.2084353 0.11308865
 [7,]                 0.2079556 0.11240367
 [8,]                 0.2066775 0.11429393
 [9,]                 0.2077010 0.11882942
[10,]                 0.2066904 0.12360364
[11,]                 0.2065279 0.12393421
[12,]                 0.2064331 0.12379935

$EXCHANGE.RATE..LAV.IN...
         TOTAL_EV GASOLINE.PRICES..INR. EXCHANGE.RATE..LAV.IN...
 [1,] 0.002151607          0.0008440543                0.9970043
 [2,] 0.003843398          0.0008408009                0.9952679
 [3,] 0.006070048          0.0232339292                0.9626529
 [4,] 0.017529491          0.0719482093                0.8862548
 [5,] 0.022749645          0.0752395729                0.8668026
```

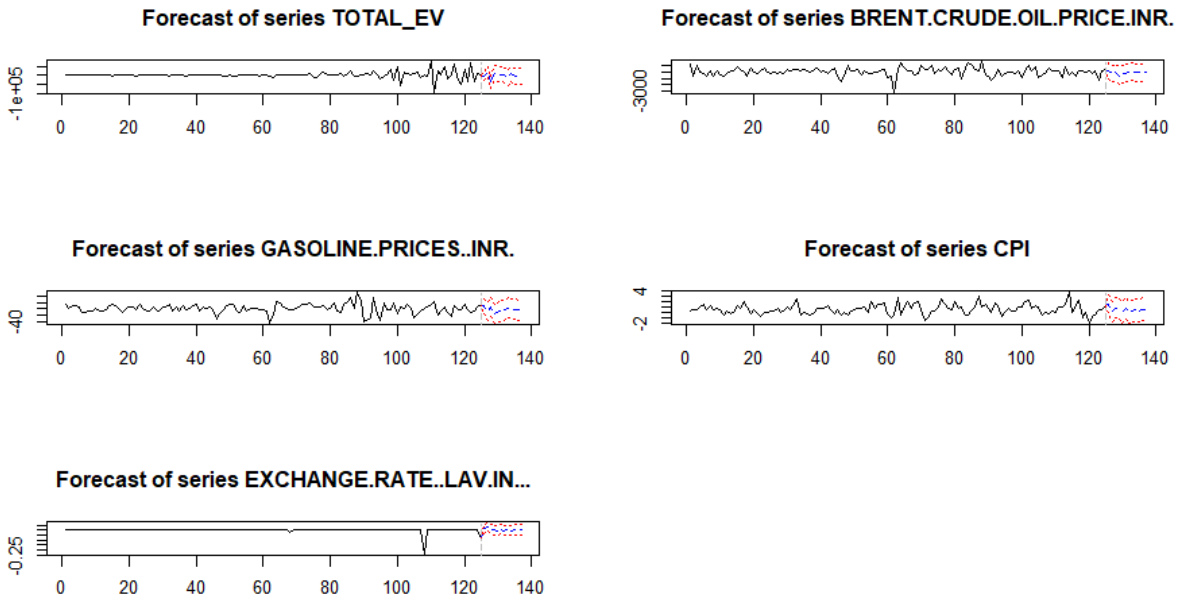
```
 [6,] 0.029067961          0.0741625018               0.8527006
 [7,] 0.029495554          0.0731382714               0.8435203
 [8,] 0.033138389          0.0727219732               0.8377027
 [9,] 0.035385909          0.0740506706               0.8337474
[10,] 0.035693479          0.0749960568               0.8323962
[11,] 0.035820191          0.0749334717               0.8312286
[12,] 0.035758594          0.0748702461               0.8301108
      BRENT.CRUDE.OIL.PRICE.INR.          CPI
 [1,]               0.000000e+00 0.000000e+00
 [2,]               1.278665e-05 3.516147e-05
 [3,]               7.449173e-04 7.298204e-03
 [4,]               1.692127e-02 7.346255e-03
 [5,]               2.528471e-02 9.923445e-03
 [6,]               3.027643e-02 1.379253e-02
 [7,]               3.039714e-02 2.344870e-02
 [8,]               3.137173e-02 2.506517e-02
 [9,]               3.122128e-02 2.559478e-02
[10,]               3.105677e-02 2.585749e-02
[11,]               3.131063e-02 2.670708e-02
[12,]               3.166626e-02 2.759413e-02

$BRENT.CRUDE.OIL.PRICE.INR.
        TOTAL_EV GASOLINE.PRICES..INR. EXCHANGE.RATE..LAV.IN...
 [1,] 0.02156316             0.4294271              0.005114248
 [2,] 0.02747140             0.4023543              0.004835206
 [3,] 0.03106277             0.3985840              0.004778454
 [4,] 0.03704808             0.3906730              0.004675037
 [5,] 0.03708792             0.3783441              0.021028049
 [6,] 0.04085592             0.3758466              0.020958423
 [7,] 0.04204010             0.3749596              0.020901186
 [8,] 0.04199903             0.3742217              0.021621968
 [9,] 0.04222387             0.3726304              0.021596931
[10,] 0.04217584             0.3718936              0.021651459
[11,] 0.04229585             0.3718404              0.021797215
[12,] 0.04230553             0.3718424              0.021817917
      BRENT.CRUDE.OIL.PRICE.INR.        CPI
 [1,]                  0.5438955 0.00000000
 [2,]                  0.5242138 0.04112532
 [3,]                  0.5187083 0.04686645
 [4,]                  0.5130888 0.05451514
 [5,]                  0.4972413 0.06629870
 [6,]                  0.4898177 0.07252136
 [7,]                  0.4897293 0.07236987
 [8,]                  0.4884659 0.07369144
 [9,]                  0.4874236 0.07612522
[10,]                  0.4865077 0.07777137
[11,]                  0.4862761 0.07779036
[12,]                  0.4862581 0.07777603

$CPI
         TOTAL_EV GASOLINE.PRICES..INR. EXCHANGE.RATE..LAV.IN...
 [1,] 0.009895033           0.008379383              0.003622742
 [2,] 0.026746675           0.008016781              0.007658392
 [3,] 0.027019519           0.013765560              0.012032401
 [4,] 0.026021505           0.041411945              0.011680740
 [5,] 0.028941933           0.042490477              0.011239614
 [6,] 0.033518478           0.041828766              0.011314732
```

```
 [7,] 0.032958590           0.042102154              0.017247589
 [8,] 0.032410903           0.044459910              0.021281856
 [9,] 0.033105837           0.047192103              0.021184131
[10,] 0.033094374           0.049093852              0.021214018
[11,] 0.033167036           0.049089380              0.021895068
[12,] 0.033015349           0.048941295              0.021924913
      BRENT.CRUDE.OIL.PRICE.INR.       CPI
 [1,]               9.061709e-05 0.9780122
 [2,]               6.697282e-03 0.9508809
 [3,]               6.851671e-03 0.9403308
 [4,]               7.520329e-03 0.9133655
 [5,]               1.464832e-02 0.9026797
 [6,]               2.518260e-02 0.8881554
 [7,]               2.475594e-02 0.8829357
 [8,]               2.480561e-02 0.8770417
 [9,]               2.544451e-02 0.8730734
[10,]               2.609882e-02 0.8704989
[11,]               2.626942e-02 0.8695791
[12,]               2.701198e-02 0.8691065
```

```
> fevd_median <- apply(fevd_results$fevd, 2:4, median)
> print(dim(fevd_median))
[1]  5 12  5
> plot_bvar_fevd_bw_dots(fevd_median, var_names, horizon = 12)
```



Forecast:

```
> var_model <- VAR(ts_data_diff, p = 4, exogen = dummy_matrix[-1, ])
> var_forecast <- predict(var_model, n.ahead = forecast_horizon,
```

```
+ dumvar = matrix(0, nrow = forecast_horizon, ncol = ncol(dummy_matrix)))
> plot(var_forecast)
```

